

Detecting and Investigating Hospital-Onset *Pseudomonas aeruginosa* Outbreaks with Machine Learning on Electronic Health Record Data



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INTRODUCTION

BACKGROUND

- Pseudomonas aeruginosa* is a common nosocomial pathogen capable of multidrug resistance that can cause life-threatening infections and represents an emerging global public health threat.
- Outbreak detection for these pathogens has traditionally relied on manual surveillance methods that are labor-intensive, inconsistent, and often delayed, limiting timely intervention at the bedside.
- Electronic health records (EHR), paired with modern computational approaches, could facilitate more rapid outbreak detection and targeted interventions.

OBJECTIVE:

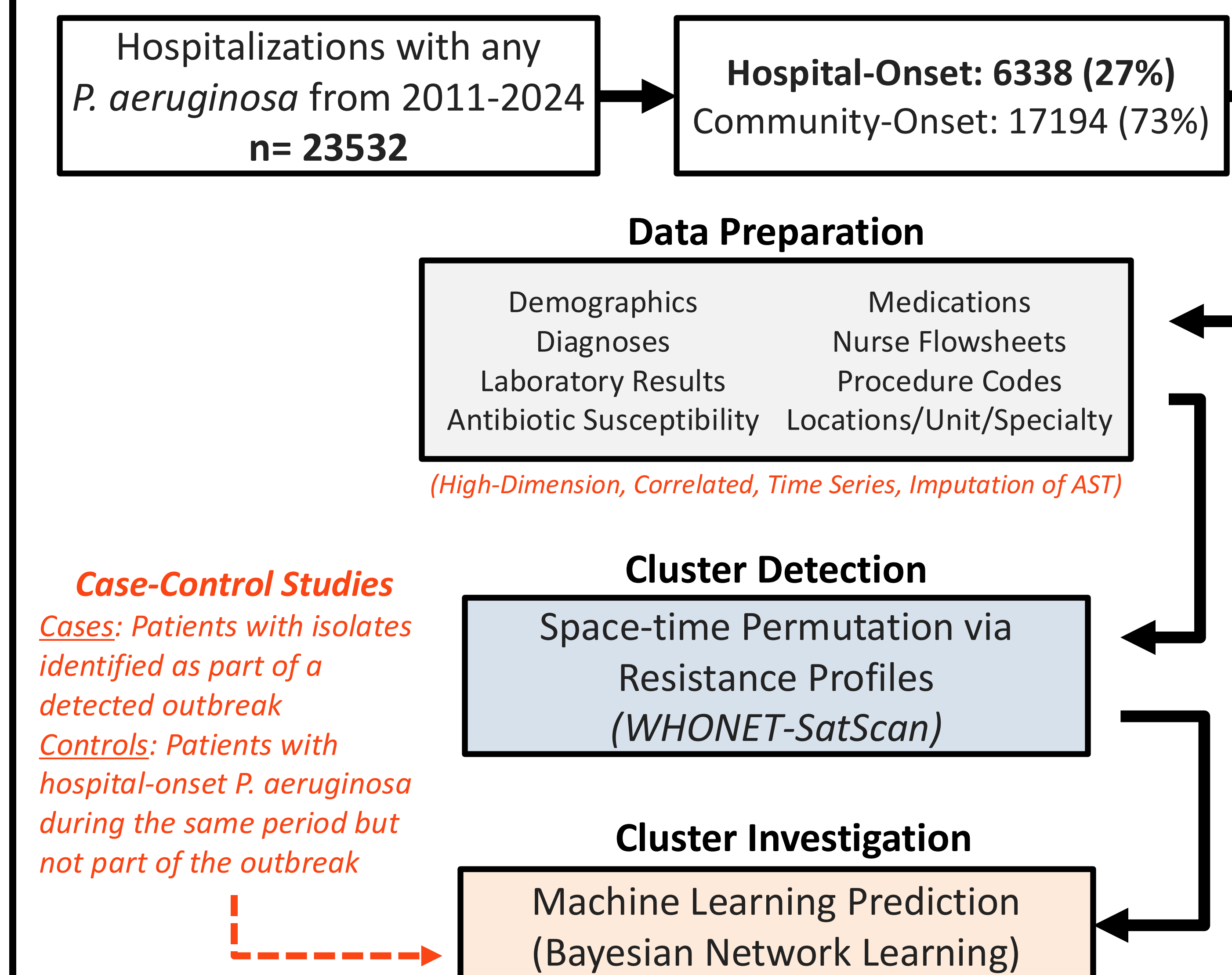
- To use routinely collected electronic health record data with machine learning to detect hospital-onset *Pseudomonas aeruginosa* outbreaks and identify potential outbreak-specific factors driving their spread.

METHODS

STUDY DESIGN

- Retrospective, population-based cohort study consisting of hospital-onset *P. aeruginosa* cases from 2011 to 2024 using EHRs (UF IDR) (IRB201900652).
- Eligibility:** Hospital-onset *P. aeruginosa* isolates were defined as culture-positive samples obtained more than two days after admission. Isolates from the same patient were included if collected > 30 days between previous isolate or change in antibiotic susceptibility interpretation.

STUDY SCHEMA



RESULTS OVERVIEW: TABLES & FIGURES

Table 1: Antibigram of Hospital-Onset *P. aeruginosa*

Antimicrobial	R% (95%CI)	
<i>Cefepime</i>	27% (26-28)	Hospital-Onset were: Odds Ratio (95%CI) MDR: 1.4 (1.3-1.5) XDR: 2.3 (2.0-2.7) Compared to Community-Onset
<i>Piperacillin/Tazobactam</i>	18% (17-20)	
<i>Ciprofloxacin</i>	23% (21-24)	
<i>Aztreonam</i>	37% (34-40)	
<i>Meropenem</i>	22% (20-23)	
<i>Tobramycin</i>	5% (5-6)	

Table 2: Summary of clusters detected using space-time permutation

Cluster Resistance Description	Cluster Date
<i>Cephalosporin, Fluoroquinolone</i>	Q3 2011
<i>Aminoglycoside</i>	Q2 2011
<i>Fluoroquinolone</i>	Q3 2011
<i>Fluoroquinolone, Carbapenem</i>	Q3 2011
<i>Cephalosporin, Fluoroquinolone</i>	Q4 2011
<i>Fluoroquinolone</i>	Q4 2011
<i>Fluoroquinolone</i>	Q2 2012
<i>Extensively Drug-Resistant</i>	Q2 2019
<i>Aminoglycoside</i>	Q4 2023
<i>Cephalosporin</i>	Q4 2023
<i>Fluoroquinolone</i>	Q2 2024

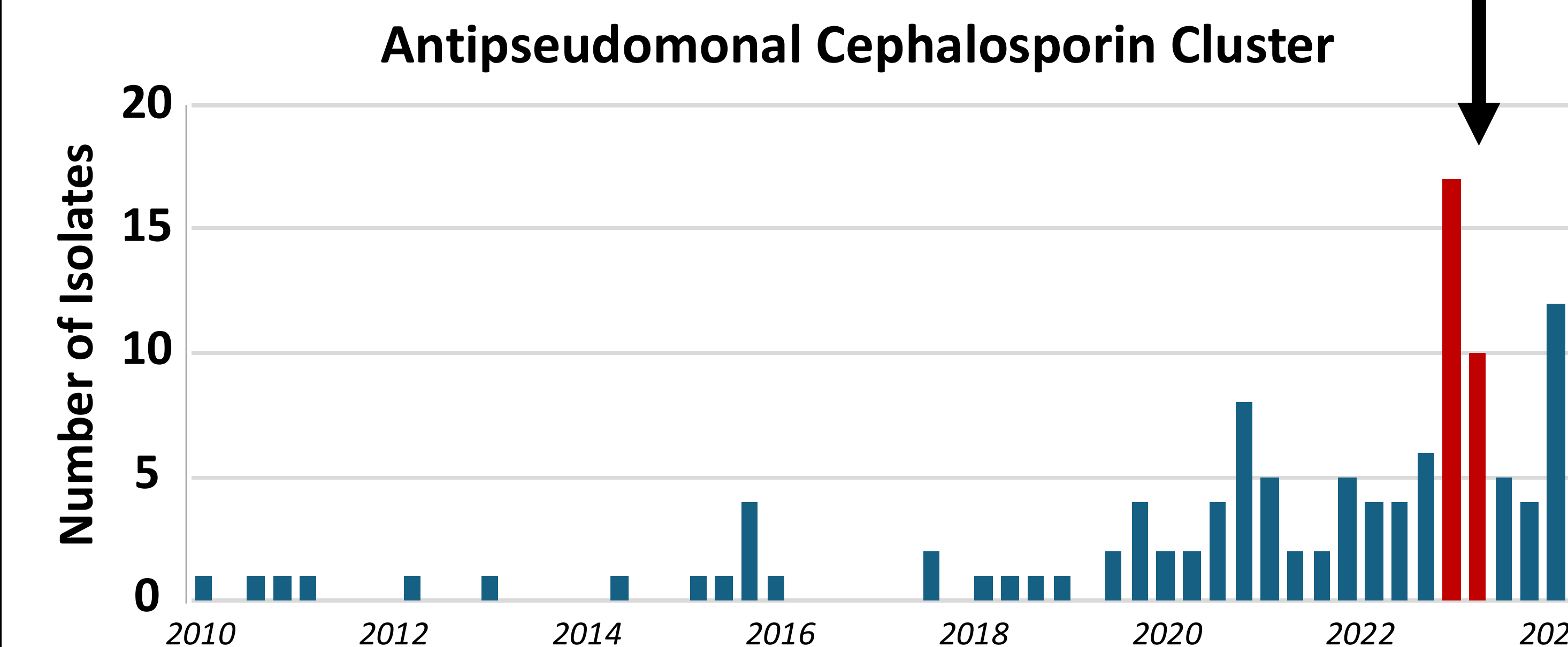
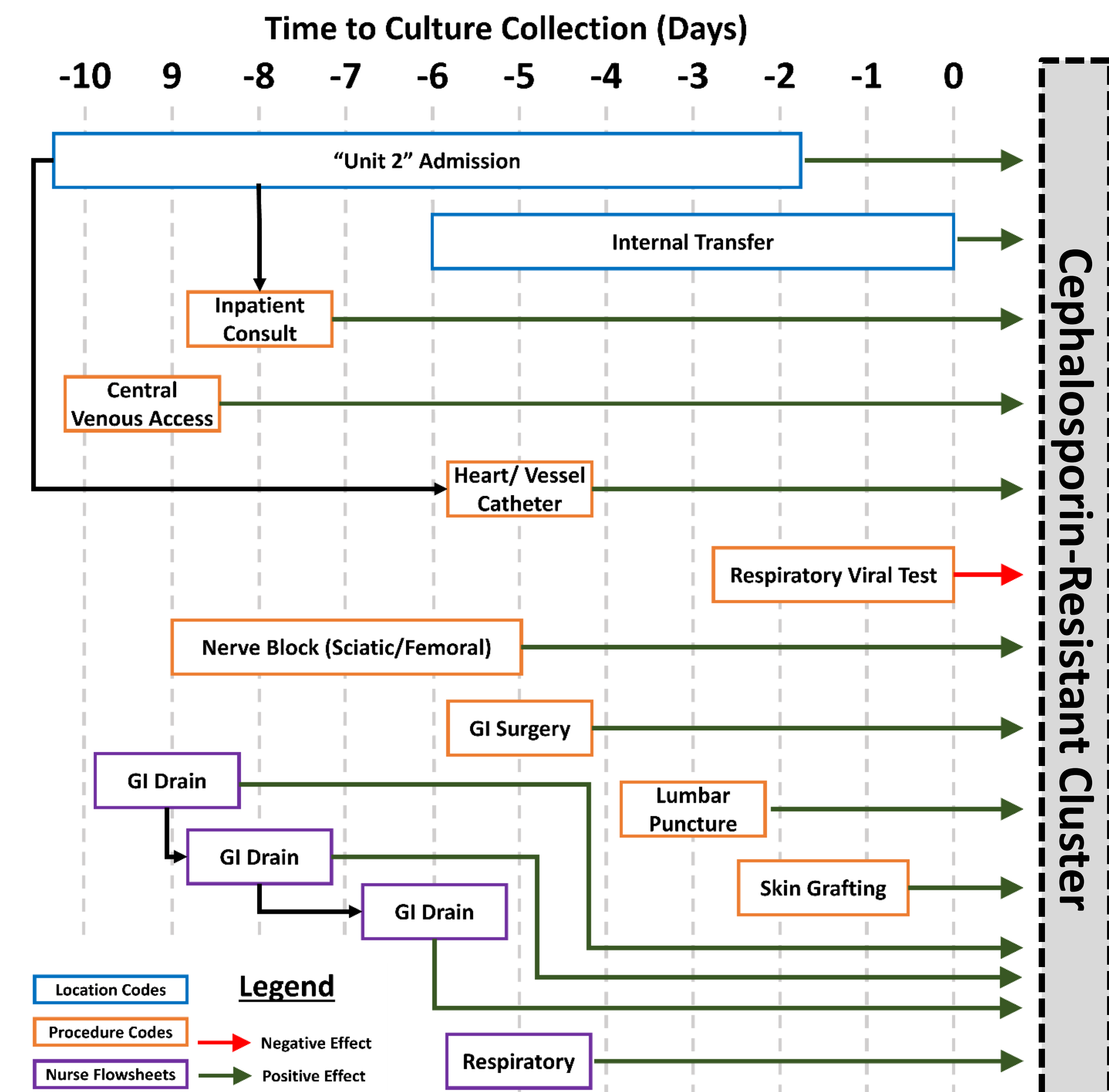


Figure 2: Machine learning model predictions for cluster membership

Cephalosporin Cluster Cases: n=36 Controls: n=360	ML Model Random Forrest Boosted Logistic Support Vector Machine Bayesian Learning	AUC 0.64 0.56 0.62 0.75
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10320 Features Screened

Figure 3: Bayesian structure model illustrating features directly associated with cephalosporin-resistant cluster membership



CONCLUSION

- Routinely collected electronic health record data combined with machine learning can detect hospital-onset *Pseudomonas aeruginosa* outbreaks.
- This approach identifies clinically plausible risk factors for transmission, including invasive procedures, device use, and unit location.
- Leveraging electronic health records for automated outbreak detection provides new evidence to inform future infection surveillance strategies.

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